

Deterministic Theories

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Abstract

Determinism is (roughly) the thesis that the past determines the future. But efforts to define it precisely have exposed deep methodological disagreements. Standard possible-worlds formulations of determinism presuppose an “agreement” relation between worlds, but this relation can be understood in multiple ways, none of which is particularly clear. We critically examine the proliferation of definitions of determinism in the recent literature, arguing that these definitions fail to deliver clear verdicts about actual scientific theories. We advocate a return to a formal approach, in the logical tradition of Carnap, that treats determinism as a property of scientific theories, rather than an elusive metaphysical doctrine.

We highlight two key distinctions: (1) the difference between qualitative and “full” determinism, as emphasized in recent discussions of physics and metaphysics, and (2) the distinction between weak and strong formal conditions on the uniqueness of world extensions. We argue that defining determinism in terms of metaphysical notions such as haecceities is unhelpful, whereas rigorous formal criteria such as Belot’s D1 and D3 offer a tractable and scientifically relevant account. By clarifying what it means for a theory to be deterministic, we set the stage for a fruitful interaction between physics and metaphysics.

1 Determinism: Capturing the intuition

The thesis of determinism seems easy enough to state:

Determinism: The past determines the future.

Unfortunately, stated this way, the thesis is uninformative, because the term “determines” on the right is just a variant of the term we are trying to define. Can we do better than this?

For many philosophers over the past fifty years, that the solutions has been to invoke possible worlds to sharpen the thesis:¹

Determinism: For any two possible worlds W, W' , if W and W' agree on the past, then W and W' agree on the future.

This possible-worlds definition gives a quasi-mathematical gloss to the word “determines”: there is a well-defined function from past segments to possible worlds. It seems like we have made good progress already in cashing out the notion of determinism. We have replaced the opaque word “determines” with what appears to be a statement about existing things and a clear relation (“agree”) between them.

But there is a problem. The relation “agree” has turned out to be a point of great contention among philosophers. To illustrate with a simple example:

(Q) Let W be a world with one particle on the left and an identical particle on the right. Let W' be the world in which the two particles are swapped. Do W and W' agree?

Indecision about Q can infect our judgment about whether determinism holds in concrete examples. For example, we can imagine that W and W' are exactly identical at all times leading up to now. If W and W' are judged to agree at the present time, then determinism holds; if W and W' are judged not to agree at the present time, then determinism fails. It seems that the fairly clear notion of determinism has become clouded by esoteric questions about what it means for possible worlds to agree.

One’s initial reaction might be that these kinds of examples only show that the notion of determinism was less clear than we imagined it was. But what are we to say about the fact that scientists seem confident in their

¹Of course, possible worlds talk is most commonly associated with Kripke (1972) and, in a different way, Lewis (1973). See Loux (1979) for a helpful early anthology showing how these ideas spread through metaphysics and epistemology in the early 1970s. It enters the philosophy of physics, and especially the analysis of determinism, most prominently with Earman (1986). Much more recently, there has been a movement away from “modality-first” approaches to metaphysics (Dorr, Hawthorne, and Yli-Vakkuri, 2021; Fritz, 2023; Sider, 2020), including among philosophers of physics (Cudek and Arana Segura, 2025). Nonetheless, the argot persists.

judgments about whether physical theories are deterministic? For example, there is a genuine difference between deterministic and stochastic equations of motion. Or, perhaps more controversially, but still to the point: it seems quite clear that there is a difference vis-a-vis determinism between quantum mechanics and the theories that went before it. Scientists appear to be able make these distinctions reliably without having a theory of when possible worlds “agree”. What is behind those judgments?

Our basic claim in this paper is that whether or not a physical theory is deterministic can be seen by analyzing the mathematical structure of the theory. Talk of possible worlds is helpful only insofar as it sometimes functions as a substitute for talk of models of a theory; for most purposes, it introduces more problems than it solves. And we do not need a different metaphysical framework to replace Lewisian possible worlds, either. Mathematics is perfectly adequate for this task.

Our arguments run against the grain of the past three decades of work on determinism in physics (and metaphysics), which has neared consensus that there can be no adequate “formal” definition of determinism. A theory, it is said, can only be deterministic or indeterministic under an “interpretation”. This is mistaken, at least as usually intended, both because it relies on confused ideas about “interpretation” and because a perfectly adequate formal criterion of determinism is already available — and has been for thirty years. We will argue that challenges for this criterion that have previously been discussed do not show that “interpretation” is needed; rather, they highlight the importance of precisely formulating a theory, as small differences in what we take a theory to be can lead to different judgments about whether that theory is deterministic.

2 The fate of formal definitions

The young Bertrand Russell frequently contrasted the clarity of nineteenth century mathematics with the opacity of Hegelian metaphysics. He pointed out that mathematicians had provided sharp definitions of the traditional number systems as well as general concepts such as infinite sets and continuous functions.

What is infinity? If any philosopher had been asked for a definition of infinity, he might have produced some unintelligible rigmarole, but he would certainly not have been able to give a

definition that had any meaning at all. Twenty years ago, roughly speaking, Dedekind and Cantor asked this question, and, what is more remarkable, they answered it. They found, that is to say, a perfectly precise definition of an infinite number or an infinite collection of things. (Russell, 1901, p 92)

Russell then proposed that philosophers should model their method on that of the nineteenth century mathematicians. Obviously, Russell's vision was a shaping force in analytic philosophy, beginning with Carnap's attempts to "explicate" the concepts of the natural sciences (see Leitgeb and Carus, 2020).

In *Logical Syntax of Language*, Carnap suggests that some deep and murky metaphysical theses correspond to precise statements about the structure of scientific theories. In particular, with regard to determinism, he says:

The opposition between the determinism of classical physics and the probability determination of quantum physics concerns a syntactical difference in the system of natural laws, that is, of the P-rules of the physical language. (Carnap, 1937, p 307)

More concretely, Carnap suggests that the metaphysical doctrine:

Every process is univocally determined by its causes

corresponds to a formal property of a scientific theory:

For every particular physical sentence φ , there is for any time coordinate t , which has a smaller value than the time coordinate which occurs in φ , a class Γ of particular sentences with t as time coordinate, such that φ is a P-consequence of Γ .

The formal property described here may seem more opaque than the original metaphysical doctrine. Nonetheless, unlike the original metaphysical doctrine, whether a theory is deterministic in the latter sense is something that could be checked by a mathematician, so long as the relevant theory has been specified in a mathematically clear fashion.

Even after Carnap's methods fell under attack by Quine, some notable philosophers continued to think of determinism as a formal property of scientific theories. J.J.C. Smart claims that:

A perfectly precise meaning can be given to saying that certain theories are deterministic or indeterministic (for example that Newtonian mechanics is deterministic, quantum mechanics indeterministic), but our talk about actual events in the world as being determined or otherwise may be little more than a reflection of our faith in prevailing types of physical theory. (Smart, 1961, p 294)

But change was in the wind, and by the 1970s, one begins to find a different kind of definition of determinism. The most influential of these “metaphysical” definitions of determinism is from David Lewis:²

A system of laws of nature is Deterministic iff no two divergent worlds both conform perfectly to the laws of that system. Second, a world is Deterministic iff its laws comprise a Deterministic system. Third, Determinism is the thesis that our world is Deterministic. (Lewis, 1983, p 360)

Lewis’ definition was put front and center in philosophy of physics by Earman (1986), and it has ever-since served as the backdrop for the “hole argument” in General Relativity (see Butterfield, 1989; Pooley, 2021). More generally, Lewis’ diverging-worlds conception of determinism informs a wide range of discussions in analytic philosophy.

What was originally an inclination to do philosophy in a different way (than the logical positivists) soon developed into theoretical arguments against formal approaches. For example, Belot (1995a) argues that determinism cannot possibly be defined in anything like the way that Carnap and Smart proposed, because it is not a *formal* property of (mathematical) theories:

The first point that I would like to make is that determinism cannot be a *formal* property of physical theories. (Belot, 1995a, p 88, emphasis in original)

Belot’s position here is part of a trend among analytic metaphysicians and metaphysically-oriented philosophers of science away from formal definitions. For example, regarding formal definitions of equivalence of theories, Sider

²Lewis credits Montague (1962) with a similar definition, but notes that Montague does not focus on metaphysical issues. We won’t discuss Montague’s paper in detail, but we suggest that his argument against a “syntactic” definition of determinism is unconvincing, and is based on a false dilemma between syntactic and semantic definitions.

claims that, “the purely formal approach is a nonstarter” (2020, p 180), and, “purely formal accounts fail because they entirely neglect meaning” (2020, p 181). A similar complaint against formal definitions of theoretical equivalence is voiced by Kevin Coffey:

The challenges posed by the two puzzles are not unique to formal approaches, but I think we should be particularly pessimistic about the prospects of formal approaches meeting those challenges. (Coffey, 2014, p 834)

And Trevor Teitel argues that formal definitions are of little interest for philosophical investigations:

I will investigate various views one might hold about the non-mathematical significance of these formal criteria, and argue that none is tenable. My tentative conclusion is that formal criteria are of limited non-mathematical interest. (Teitel, 2021, p 4120)

What justifies this widespread rejection of formal methods? Sider, Coffey, and Teitel make different arguments from Belot, and from one another. But they also put great weight on the importance of *interpretation* or, as Sider says, *meaning*. We will focus on Belot’s argument against formal definitions of determinism. When we see why that argument fails, it will be clear why we would reject other arguments for the same kind of view. And then, with formal methods rehabilitated, we will turn to a formal analysis of other arguments in the literature on determinism.

3 Interpretation is a formal matter

Belot’s argument for the claim that determinism cannot be a formal property of theories involves an example. He presents a set of equations — Maxwell’s equations, describing classical electromagnetism, written in a particular way — that he claims are standardly understood to be part of a deterministic theory, but which may yet also be part of an indeterministic one. The difference, he says, comes down to interpretation.

This completes the argument: determinism cannot be a formal property of theories, because the same theory may be deterministic or indeterministic, depending on how it is interpreted. (Belot, 1995a, p. 88)

The argument as stated has a suppressed premise: that interpretation is not a formal matter. But we deny this premise, twice over. First, the claim that interpretation is not a formal matter takes interpretation to involve a re-negotiation of the relationship between language and reality. Second, and relatedly, the claim that interpretation is not a formal matter fails to see that interpretation of a theory very often calls for a formal precisification of that theory.

We can make both points by describing Belot’s example in more detail. As he set it up, it concerns two theories. One of these theories is classical electromagnetism, written in terms of a scalar (electric) potential and a vector potential, and satisfying Maxwell’s equations. The second theory concerns the properties of a “plenum of point particles” called “blips” (88). Blips are represented by two fields on (Newtonian) spacetime: a scalar field φ representing the charge of the blip(s) at each point of spacetime; and a vector field A representing their velocity at each point. The blip theory is remarkable because it describes the motion of blips using just two equations—equations that happen to be syntactically identical to Maxwell’s equations written for a scalar potential φ and vector potential A .³ Belot argues the first of these theories is deterministic, standardly (and properly) construed. But the second is indeterministic. The reason is that a specification of the charges and velocities of blips throughout space at a time cannot uniquely determine the charges and velocities of blips at other times. Given any solutions to the blip equations, one can always apply any of a family of transformations to produce another solution that agrees on the initial specification. This is so even though they involve the same equations as Maxwell’s theory.

How could it be that one of these theories is deterministic, but the other is not? Cognoscenti may guess the answer, but we will postpone giving it. Instead, consider what Belot takes to be at issue. For him, the difference between the theory is a matter of how they are interpreted. What does he mean by “interpretation”? He poses and answers the question directly.

What *is* an interpretation of a theory? An interpretation of a theory is a story that you tell about the theory. . . . An interpretation is a correspondence between bits of models of the theory and bits of physical situations: between initial value constraints, variables and differential equations on the one hand, and instantaneous states, physical entities, properties, relations, etc., and

³See Belot (1995a, p. 88) for details, and a fine example of his inimitable style.

laws of nature on the other. (Belot, 1995a, p 92)

This passage suggests a widespread view in late twentieth century analytic philosophy, which is that to interpret something like a physical theory is to apply some “semantic glue”, thereby attaching symbols to their worldly referents. But how does this metaphor actually work? Belot’s articulation should give us pause, because it involves an odd mix of two very different things. The first is the mathematical theory of interpretation, developed by Tarski et al. six decades before, which explicates an interpretation of a formal language as an assignment of set-theoretic extensions to meaningless symbols. Belot’s talk of the “models” of a theory invokes this tradition.

But Belot also motions towards a much woolier, and apparently older, sense of “interpretation” when he speaks of a correspondence between models and the world. In doing so, Belot has re-doubled the notion of interpretation. Recall that in its Tarskian sense, an interpretation of a theory *is* a model of that theory. But now Belot speaks of interpretation as, “a correspondence between bits of models of a theory and bits of physical situations”. In this case, the interpretations (models) of a theory are being interpreted. But what does that mean? How would we produce or exhibit a “correspondence between bits of models of the theory and bits of physical situations”?

Belot does not say how this is supposed to go in general. And when it comes time for him to describe two theories that differ (only?) in interpretation, what we find is not a “correspondence between bits of models of the theory and bits of physical situations”, but rather some *words*, further elaborating some formal aspects of two theories he wishes us to understand. The exercise is all carried out on paper, by sketching a correspondence between bits of models of the theory and yet another kind of symbol: words in natural language. But this is just to layer (formal, symbolic) “interpretation” on “interpretation”—that is, to do the “redoubling” we complained about above. And on reflection, given that he has written a paper, it is hard to see what his sense of interpretation could amount to aside from creating yet other models, and associating bits of the old models to bits of the new models. So in fact, though he motions at something different, what he exhibits as interpretation does not give us some new way of crossing the word-world barrier; it merely connects things that lie on the word side, the formal side, to other things on that side.

One might well object here: how does *any* of our linguistic or symbolic practice have meaning, if all of our “interpretation” consists of layering models

on models and words upon words? Surely, one might say, we do sometimes interpret in the woolier, and perhaps deeper, sense — and at least some of our words have meaning. Of course we accept this. But we suggest this line of thought just leads to a dual perspective on our position, entirely consistent with what we have said thus far. From this perspective, *all* of the formal, mathematical, linguistic, and symbolic structures that we employ, in ordinary life, in philosophy, and in science, have meaning, at least to some extent, because of the way they are embedded in our broader cultural and cognitive processes. To put the point in terms of the metaphor noted above: there is no point in the paper in which Belot *applies* semantic glue. The glue — or rather, whatever processes allow any of our symbolic activity to play the role it does in reasoning, action, and communication — is already there.

Dewar (2023) puts the point nicely.

We don’t begin our analysis of scientific theories by taking some mysterious equations carved on stone tablets and puzzling out what they might mean: theories are born as bearing all kinds of semantic or interpretational relationships to our broader representational practice. ... [T]he problem is not how to comprehend an alien practice, but how to fully understand a practice which we already — at least to some extent — inhabit. (18)

Again, we will not attempt to give a story of how meaning works. Our point is that there is no line to draw between “merely” or “purely” formal methods or structures, between truly “meaningless symbols” or marks on a page, and any other symbols we might try to employ to make our ideas clear or communicate them to others. The idea from the formal theory of interpretation that we are giving meaningless symbols “meaning” by mapping them to set-theoretic structures is an idealization — as should be obvious, since *both* sides of the interpretation map consist of rich but thoroughly formal theories.⁴ Likewise, interpretation in the wild consists of using meaningful symbols to guide reasoning about other meaningful symbols. It is all just more of the same.

⁴ Some readers will raise an objection at this point about the role of (literal, physical) “ur-elements” in formal semantics. Addressing this thought would take us too far afield and we will leave it to future work. For now, suffice it to say that we reject the idea that physical things can be “in” abstract ones; set theory with ur-elements is best viewed as a formal theory living within ordinary set theory, where some abstract objects have highly suggestive names.

To illustrate the point, we now return to how Belot can maintain that electromagnetism is deterministic, while blip theory is not. Belot does not fully articulate these theories in a formal way. But he describes them with sufficient clarity that one can see why they are in fact mathematically distinct — and it is precisely those mathematical differences that support his judgments about whether they are deterministic. The crucial step occurs when, in describing the theories, he explains when putatively distinct models of electromagnetism actually correspond to the same physical situations (described in different terms)—whereas for the blip theory, those same models correspond to distinct physical situations. This amounts to a disambiguation between two possible theories, given by specifying when models are *isomorphic*, i.e., when they should be seen as mathematically (and, for the purposes of analyzing the theory, semantically) equivalent. According to one disambiguation of these theories (namely, electromagnetism), there is an isomorphism between models if they are related by a certain class of transformations known as *gauge transformation*.⁵ Among the gauge transformations are all of the transformations that related “distinct” evolutions of a given initial configuration of the blip theory. Meanwhile, according to the disambiguation of electromagnetism — or rather, the blip theory — gauge transformations are not isomorphisms, and thus count as (formally) distinct by the lights of that theory. This is what supports the conclusion that electromagnetism is deterministic but the blip theory is not.

The important point for our purposes is that everything here is formal: the description of the theory — both models and the relationship of isomorphism between models — as well as the definition of determinism in the background. Belot has not escaped from the circle of formalism and into a realm of inarticulable concreta; he has merely motioned toward the power of formal tools to capture conceptual distinctions. And we thank him for this: he shows clearly that what appears to be a single theory is in fact two, depending on how one spells out the details. We can call this explication of formal structure “interpretation” if we want — as Tarski did — but in that case, interpretation is a formal matter.

Of course, we have not given a first-order theory then specified a Tarskian interpretation function. We operate instead at a level closer to ordinary

⁵See Weatherall (2016) for a discussion of this way of understanding gauge transformations in electromagnetism, along with a different way of distinguishing standard electromagnetism from its close cousins with different isomorphisms.

mathematics. But even so, it shows how the salient activity for “interpreting” our theories involves *formal* work— in this case, the formal (or mathematical) work of specifying which models are isomorphic — because a different specification of the relationship of isomorphisms gives a different theory, and it might transform a deterministic theory into an indeterministic one, or vice versa. We will presently see a more subtle example of where careful interpretation of a different kind, though still formal, can disambiguate between theories whose differences are difficult to detect through more conversational descriptions.

4 From explication to metaphysical speculation

With that defense — and explanation — of formal methods in mind, we now return to analyses of determinism downstream of Lewis. There are two directions in which Lewis’ (1983) definition of determinism has been developed. One is the quasi-mathematical approach introduced by Earman and Butterfield, but then attacked by Belot. We return to that below. First, we will consider the other direction of development. On this branch, Lewis’ definition has been suited out with various metaphysical distinctions — and especially an elusive distinction between “qualitative” and “full” agreement of possible worlds. These metaphysical adumbrations on Lewis’ definition have given rise to a significant literature which might seem, at first, to evince a rich interaction between physics and metaphysics. But as we will presently argue — bringing formal methods to bear — these appearances are misleading.

Consider Hawthorne’s (2006) proposed distinction between two senses in which the world might be deterministic.⁶

Qualitative Determinism: For all times t , there is no possible world which matches this world in its qualitative description up to t , and which has the same laws of nature as this world, but which doesn’t match this world in its total qualitative description. (239)

De Re Determinism: For all times t , there is no possible world which matches this world in its de re description up to t , and which has the same laws of nature as this world, but which doesn’t match this world in its total de re description. (239)

⁶A similar distinction can be found in (Melia, 1999), though not quite as sharply put.

Hawthorne’s distinction seems to be nearly ubiquitous in the literature. For example, Teitel (2019) distinguishes laws that are qualitatively deterministic from laws that are “fully” deterministic. Similarly, Pooley (2021) claims that what he calls “substantivalist” General Relativity is deterministic to a lower degree “Det2”,⁷ but not in the most eminent sense “Det1”.

	grade 1	grade 2
Hawthorne	qualitative	de re
Teitel	qualitative	full
Pooley	Det2	Det1

Each of these authors indicates that there is some thicker kind of determinism that is not established by standard proofs, e.g. of the initial value problem for partial differential equations.

Clearly, the terminology here has become variegated.⁸ For simplicity, we will use Teitel’s “full determinism” to gather together the different senses of the highest grade of determinism. We will argue that previous attempts to define full determinism fail: it cannot be cashed out either in terms of types of descriptions, nor in terms of some equivalence relation on possible worlds. We do not conclude, however, that qualitative determinism is the only kind of determinism worth worrying about. In fact, it is possible to capture the distinction between qualitative and full determinism, properly construed, in “purely formal” terms, viz. in terms of uniqueness of extensions of isomorphisms. Appearances to the contrary arise due to confusion about (formal) interpretation and imprecision in the individuation of theories.

For possible worlds W, W' , let us write $W \sim_q W'$ if W and W' match in qualitative description, in Hawthorne’s sense (i.e. are “qualitatively iden-

⁷To be sure, Pooley (2021, §5) also argues that an anti-haecceitist – such as himself (c.f. Pooley, 2006, pp.99-103) – would maintain that Det1 and Det2 are strictly equivalent, though he also feels the draw of examples, such as those discussed in the main text, intended to show Det2 to be defective. We see no value to the qualifier “substantivalist” here, since what Pooley has in mind is apparently just textbook General Relativity. We will return to this point below; for now, we will drop the qualifier.

⁸Or worse, because the terminology is also not always consistently applied. For instance, Dewar (2016; 2024) proposes a more subtle distinction between “de dicto” and “de re” determinism, where his “de dicto” is clearly grade 1, but his “de re” determinism is weaker than Hawthorne’s. (Actually, it is similar to Belot’s D2 or Melia’s “Second Resolution” 1999.) Manchak, Barrett, et al. (2025) use the terminology much like Dewar. Cudek (2024) tracks the same distinctions as Manchak, Barrett, et al. (2025), but using terminology that follows the model of Butterfield (1987). And so on.

tical” in Pooley’s sense). Let’s write $W \sim_r W'$ if W and W' match in de re description, in Hawthorne’s sense (i.e. are “intrinsically identical” in Pooley’s sense). While we agree that it is possible to distinguish between grades of determinism, we will show that such distinctions cannot be based on some distinction between qualitative and de re equivalence of worlds. In the remainder of this section, we will try to give the distinction between $\sim_q$ and $\sim_r$ a run for its money — concluding that it only gives intelligible answers for theories with names for all objects.⁹ In Section 5, we propose a distinction between grades of determinism that does not rely on a distinction between qualitative and de re equivalence of worlds.

4.1 Descriptions: Qualitative versus De Re

Hawthorne and Teitel explain the distinction between $\sim_q$ and $\sim_r$ in terms of a distinction between types of propositions. Roughly speaking, $M \sim_q M'$ means that $M \models \varphi$ iff $M' \models \varphi$, for all qualitative propositions φ . And $M \sim_r M'$ means that $M \models \varphi$ iff $M' \models \varphi$, for all de re propositions φ . Can we make rigorous sense of a distinction between these two types of propositions?¹⁰

Hawthorne explains the distinction between two kinds of descriptions as follows:

The first — the *qualitative description* — says everything that can be said about the intrinsic character of that history with one exception: it cannot name individuals or otherwise encode haecceitistic information about which particular individuals are caught up in that segment of world history. The second — the *de*

⁹The term “name” is loaded in the philosophy of language. To forestall misunderstanding: when we write “name” we mean “constant symbol in a formal language”, except where it is clear from context that we are entertaining other views.

¹⁰Of course, we are hardly the first to ask this question. Indeed, there is a large literature on the distinction between qualitative and non-qualitative propositions (see e.g. Kaplan, 1975; Adams, 1979; Rosenkrantz, 1979; Dasgupta, 2009; Cowling, 2015; Hoffmann-Kolss, 2019). It seems to be widely agreed among metaphysicians that the distinction is intelligible and important, and yet that making it precise is elusive. Several of the moves we make here will be familiar to some readers, though we do not attempt a general survey of the literature. (For that, Cowling (2015) is especially helpful.) Instead, our goal is to show that it is unlikely that a sharp *mathematical* distinction between qualitative and non-qualitative properties is available — and thus, that the line of thought we are currently contemplating will not yield the sort of account we seek to defend.

re description — includes the qualitative description and, in addition, all haecceitistic, singular information. (Hawthorne, 2006, p 239)

Granted, such a distinction works fine in everyday life: in some contexts where we have a name for an individual, we can give a specific description of what properties that individual has, or we can give a general description of something that has those properties. This is a distinction between levels of generality.

And yet, for well known reasons, the distinction is not absolute. For instance, sometimes qualitative descriptions can contain singular information. As Russell himself taught us, the sentence

(D) The current president of the US is bald.

does not contain the name of an individual, but is, in some sense, about a particular individual. Similarly, Quine pointed out that a name such as “Pegasus” could be replaced by a predicate “pegasizes” (Quine, 1948). Conversely, even names may not include all singular information, at least in ordinary language.¹¹ “James Weatherall” may refer to a philosopher of physics, or to his father, or his son, or to a former All-American football player, or to a retired British Vice-Admiral. In each of those cases, further descriptive information is needed to disambiguate the reference of the name.

Perhaps more importantly, whether some description contains singular information is apparently not a fact about the description itself, but rather about the thing or things described. For example, “Some nobleman is bald” could be about no individuals, or about one, or about many. And in any of these cases, it implies the less specific proposition “Somebody is bald”. On these grounds, we are skeptical about the idea that there is some significant distinction between descriptions that encode information about individuals, and descriptions that do not.

But let’s try harder. Recall that a description is supposed to be qualitative just in case it does not name any particular individual. We will now consider several proposals for how to make this idea precise. We assume that propositions are expressed, up to logical equivalence, by sentences. Thus, we assume that if sentences are logically equivalent, then either they both name an individual, or neither of them names an individual. If that were

¹¹In first-order logic, we regiment name usage to require names to refer uniquely. But that is a choice in how we set up our semantics, and it does not apply in ordinary language.

not the case, then “ x names an individual” would not be a property of the underlying proposition, but only of some specific syntactic representation of that proposition, and that would not help in defining a notion of qualitative sameness of worlds.

Proposal 1. A sentence φ names an individual just in case φ contains a name.

We have already seen arguments against proposals of this sort, but stated in these terms it fails for even simpler reasons, since it depends on superficial features of a sentence that are not invariant under logical equivalence. Indeed, φ is logically equivalent to $\varphi \wedge (c = c)$, so by this proposal, every sentence names an individual.

Proposal 2. A sentence φ names an individual just in case there is a name c such that for any sentence φ' , if φ' is logically equivalent to φ , then φ' contains c .

This proposal would make it essentially impossible for sentences to name individuals. Indeed, one could define a predicate $\theta(x) \leftrightarrow (x = c)$, and then replace the de re sentence $p(c)$ with the definitionally equivalent $\exists!x(\theta(x) \wedge p(x))$.

Proposal 3. Given a background theory T , a predicate $\theta(x)$ names some individual relative to T just in case $T \vdash \exists!x\theta(x)$.

This condition is not strong enough. Even when $T \vdash \exists!x\theta(x)$, there can be a model M of T such that $M \models \theta(a)$, and another model M' of T such that $M' \models \neg\theta(a)$. So, $T \vdash \exists!x\theta(x)$ does not capture the idea that $\theta(x)$ names an individual.

We have tried, without any success, to give a formal (i.e. mathematical) account of the distinction between qualitative and de re propositions. Perhaps our failure is unsurprising, since the distinction between qualitative and de re descriptions was intended to be a distinction in how those descriptions relate to the world, i.e., it is meant to be a distinction in what those descriptions are *about*, whereas we have been talking only about mathematical objects (e.g. sentences of formal languages, models).

Here is another strategy. Perhaps one could simply assume that there is a relation $N(\varphi, x)$ of “naming” that can hold for propositions and concrete objects.

Proposal 4. A proposition φ is *qualitative* just in case $\neg\exists xN(\varphi, x)$.

Such a proposal has all the advantages of theft over honest toil. But to be more serious, we are still hung up on the practical concern of whether the distinction would actually help us decide whether specific scientific theories are deterministic, and saying that there is such a relation N does not tell us which sentences are qualitative and which are de re. Nor would this proposal provide any concrete guidance on the question of whether models are de re or qualitatively equivalent. Metaphysicians would be within their rights to say “there is such a relation N ”, but until they tell us how to detect that $\exists xN(\varphi, x)$ for a proposition φ , their proposal is of no interest for understanding the metaphysical implications of science.

Perhaps, you might think, we are straying too far from common sense. Surely the following claim, which one might find in a science textbook, names a concrete physical object:

(E) The perihelion of Mercury advances by approximately 43 arc-seconds per century more than what is predicted by Newtonian mechanics.

But for reasons related to our arguments in section 3, we think this rests on a linguistic confusion. Yes, the sentence involves a name in the colloquial sense. But the sentence most certainly does not involve “all singular, haecceitistic” information about the planet closest to the sun. It manages to convey the intended meaning only because of its integration with linguistic and scientific practice, our history and culture, established conventions, and so on. Put another way, the term “Mercury” does not do anything, it is people who do things. Names and sentences and propositions do not name things, it is people who name them. We may wish to include an abstract intermediary, such as a proposition, that a person uses to refer to something—and we might single out certain non-logical vocabulary as especially well-suited for playing a referential role in our languages. But without the person in the middle, the proposition does not stand in an intrinsic relation to any concrete thing.

We find that we have entered a wild and unfamiliar metaphysical territory, far outside of our comfort zone. We have no carefully worked out view of the metaphysics of propositions, and we do not intend to take a stance on these issues. More power to them who believe that they have a theory of propositions that will shed further light on the elusive distinction between qualitative and full equivalence of possible worlds.

What we do intend to take a stance on is that some definitions are more useful than others. For example, what does it mean to say that two sentences are synonymous? As Quine pointed out, it is perfectly correct, and perfectly useless, to say that two sentences are synonymous if they express the same proposition. The problem is that nobody has ever figured out that two sentences are synonymous by comparing each of them with a proposition. In the same way, nobody will ever determine if a proposition φ is de re by checking $\exists x N(\varphi, x)$.

4.2 Equivalence relations on models

The attempt to distinguish qualitative from de re equivalence of worlds via a distinction in kinds of descriptions strikes us as a dead end, at least from the perspective of trying to assess when a physical theory is deterministic. Let's approach the issue from a different direction. We are interested in two proposed senses of when two worlds "agree". Perhaps we can capture this "agreement of worlds" more directly via models of a theory.

Hawthorne supplies a motivating example of worlds that are supposed to be qualitatively equivalent but de re inequivalent.

Consider a symmetrical world where there is a pair of qualitatively identical ships, one in each symmetrical half. Suppose the laws dictated that exactly one of the ships would sink, but left it undetermined which. Qualitative determinism might still hold of such a world, since the qualitative description of a world in which one ship sank need not depart in any way from the qualitative description of a world in which the other did. (Hawthorne, 2006, p 243)

We share Hawthorne's intuition about which counterfactual statements are true in this situation. In particular, we think it is true that no matter which ship sinks, it could have been the case that the other ship sank. And insofar as we can specify the initial state of the world without specifying the truth-value of future contingents, the initial state of the world does not determine the final state. So there is clearly a sense in which this example manifests a kind of indeterminism.

The key here is the phrase "the other ship". There are two ships to begin with, and we can stipulate that one of them be called "Lefty" and the other "Righty". There is nothing in the description that tells us which ship sinks

— it could be Lefty, or it could be Righty. If Lefty does in fact sink, then it is still true that it could have been Righty that sank. Indeed, there are *two* distinct possible final states of affairs, both of which are compatible with the initial state of affairs.

All of this common sense talk can be shored up by formal model theory. Consider a theory T that says that there are exactly two objects, two definite descriptions $L(x)$ and $R(x)$, and a single predicate $P(x)$ to indicate which ship sinks at the latter time. While a model M of T consists of a domain with two elements, and extensions for L, R and P , an initial segment of this model simply omits the extension of P — i.e. it does not specify which ship sinks. There is then a clear sense in which a single initial segment can belong to two distinct models. For example, the initial segment where a is Lefty and b is Righty belongs both to a model where Lefty sinks, and to a model where Righty sinks. To be more precise, a model where Lefty sinks is one where $\exists x(L(x) \wedge P(x))$ is true, while a model where Righty sinks is one where $\exists x(R(x) \wedge P(x))$ is true. Since models are isomorphic only if they satisfy the same sentences, a model where Lefty sinks is non-isomorphic to a model where Righty sinks.

There is also a clear sense in which any two models of T are qualitatively equivalent. The language of T is the union of two sublanguages $\{P\}$ and $\{L, R\}$, where the latter can be considered as the “haecceitistic” vocabulary. In this case, we can say that M is qualitatively equivalent to M' just in case $M|_P$ is isomorphic to $M'|_P$. It then follows that any two models of T are qualitatively equivalent — and this underwrites the intuition that the process is qualitatively deterministic. We can then elevate this idea into a proposal for how to determine when models of a theory represent equivalent worlds.

Proposal 5. For models M, M' of T , we say that M and M' are *qualitatively equivalent* if M and M' are isomorphic qua structures for the qualitative sub-language. We say that M and M' are *de re equivalent* if M and M' are isomorphic qua structures for the full language.

With this distinction in place, we can then define:

Full Determinism. Let T be $(\Sigma_q \cup \Sigma_r)$ -theory. Then T is deterministic just in case for any models M, M' of T , and for any initial segments U of M and U' of M' , if U and U' are de re equivalent, then M and M' are de re equivalent.

This definition gives reasonable answers for the simple kinds of examples we have considered. In particular, it explains the sense in which Hawthorne’s ship example fails to be fully deterministic.

Suppose now that we take Proposal 5 and try to apply it to a theory that purports to describe the actual world, e.g. General Relativity. Here we face an obstacle: General Relativity does not provide names for spacetime points. Can this be overcome? In fact, something similar could be said of Hawthorne’s theory. Hawthorne said “consider a pair of qualitatively identical ships”, and then *we* proceeded to name these ships “Lefty” and “Righty”. We might say, then, that Hawthorne’s original theory does not have names, in the technical sense of constant symbols assigned to domain elements, but we can extend that theory in a way that introduces names and breaks the symmetry between the two ships. In doing so we introduce a new theory, which is such that every model of the original “theory” (i.e. “there are two ships and one sinks”) corresponds to two models of the new name-enriched theory.

Imagine that you were learning General Relativity and the professor said, “a model of Einstein’s Field Equations consists of a Lorentzian manifold . . .”, and then you replied “I’m going to add a name ‘Bob’, so that in each model, ‘Bob’ refers to some particular spacetime point.” We have nothing against adding vocabulary to a theory, but we insist that the result will typically be a *different* theory. In fact, GR+Bob has a predicate, ‘ $x = \text{Bob}$ ’, that is nomically isolated, i.e. it has no lawlike connections to any other predicates or relations in the theory; and adding nomically isolated predicates to a theory will necessarily result in an indeterministic theory. For example, if N is the theory of inertial particle motion in Newtonian absolute spacetime, then adding two one-place predicates A and B will result in an indeterministic theory — since Newtonian mechanics does not say whether a particle that satisfies A at one time will satisfy B at some later time.

Now, one might think that the issue in these examples is not really about adding names to a theory, because somehow the resources needed to talk about individuals are already available without any extension. Indeed, in typical discussions of the hole argument in general relativity, the labels for spacetime points are not taken to be part of the syntax of the theory, but are assumed to be added by us, after we choose a specific spacetime M . i.e. we don’t assume that there is some name ‘Bob’ that refers to a point in every spacetime. Instead, once we are given a spacetime M , we note that it is a non-empty set, and we say “consider some $p \in M$ ”. We then note that

p has some property φ in M , but there is a mathematical construction that generates another spacetime M' based on the same set, and in which p does not have property φ . The thought then is that M and M' represent de re inequivalent worlds.

We could look at Hawthorne’s example in the same fashion. Don’t introduce names ‘Lefty’ and ‘Righty’ at the start, but just write down a model $M = \langle \{a, b\}, a \rangle$ for “the theory of two ships, where one sinks”. Clearly $M' = \langle \{a, b\}, b \rangle$ is also a model for this theory, and we could imitate the hole argument by constructing M' from M . The fact that $M \neq M'$ as sets seems then to underwrite the claim that there are two de re inequivalent worlds.

But what is the basis for saying that M and M' represent de re inequivalent worlds? Consider a possible world with a pair of objects, one of which is P . We can name the thing that is P in various ways: we could name it a , or we could name it b . So why not take $\langle \{a, b\}, a \rangle$ and $\langle \{b, a\}, b \rangle$ to be two different representations of the exact same world? Nor will it help here to say: “but a and b are not names of things, they are the things themselves”. That is just a bit of nonsense, for reasons already discussed.¹²

The claim that M and M' represent de re inequivalent possibilities is less solid than it might initially seem. How might it be justified? The obvious answer here is that M and M' are set-theoretically distinct: M assigns $\{a\}$ to P , while M' assigns $\{b\}$ to P . This suggests the following proposal:

Proposal 6. Models M and M' represent de re equivalent worlds only if $M = M'$ in the sense of Zermelo-Fraenkel set theory.

We ourselves are inclined to think that set-theoretic identity of models is of no significance here; we think that isomorphism of models is the only significant standard of sameness.¹³ But we do not need to convince you of that stronger position. It is enough to point out that Proposal 6 supplies far

¹²Recall fn. 4 and the surrounding discussion.

¹³Reasons for thinking this in a related context are given by Weatherall, 2018 and Bradley and Weatherall, 2022; other reasons, related to the arguments in section 3, are discussed below. Ultimately, the issue is that insisting on set-theoretic identity of models involves layering interpretation on interpretation in a way that fundamentally confuses what the theory expresses. Perhaps ironically, given the subsequent literature, while Kaplan (1975) both introduces “haecceitism” to contemporary philosophy and also suggests, in the somewhat different context of Kripkean modal semantics, that set theoretic identity is a reasonable way to represent haecceitistic facts, he *also* suggests that the anti-haecceitistic philosopher has an easy way to avoid taking set theoretic identity to have

too many possible worlds, viz. as many as there are sets. For example, when Hawthorne begins describing his example by saying “consider two ships”, we are left wondering whether the first ship is a or $\{a\}$ or $\{\{a\}\}$ or $\dots$. We cannot speak for metaphysicians, but no physicist would say that it matters which element a we use to represent the first ship, as long as we use a different element b to represent the second ship. In fact, the reason we write “ a ” and “ b ” as opposed to something like “ $\{\emptyset\}$ ” and “ $\{\emptyset, \{\emptyset\}\}$ ” is because we have no good reason to choose one set over another to stand in for our ships. We are not really supposed to consider the first ship to be a *specific* object in a model of ZF set theory, but as a generic element of a two-point set. More to the point, standard practice in mathematical physics allows that the same possibility be represented by distinct sets. Nobody ever asks whether a spacetime M contains $\{\emptyset\}$ — because that would be a representational artifact.

Besides, even if a spacetime model really is a set, we should be skeptical that that should make a difference to whether general relativity is deterministic. After all, the working physicist never cares which set it is, as long as it has the right structure, and this structure is specified by the syntax of the scientific theory — not by set theory. Therefore, set-theoretic equality of models should not be viewed as a necessary condition for de re equivalence of the corresponding worlds, since set theoretic equality is irrelevant to any sort of physical equivalence. Proposal 6 only sounds promising until one thinks about how many different sets there are, and about how irrelevant it is which set we use.

We’ve been trying to figure out when models of a theory represent de re equivalent possibilities. The first proposal along these lines, proposal 5, was that we add names up front (enriching the syntax of the theory), and declare that models represent de re equivalent possibilities only if they are isomorphic relative to the name-enriched language. But that proposal won’t help determine whether the original theory is deterministic, since adding names to a theory results in a new theory. The second proposal, proposal 6 attempts to cash out de re equivalence in terms of set-theoretic identity of models. Like the first proposal, this second proposal tacitly replaces the orig-

this significance, viz. by introducing suitable isomorphisms. All of the theories we consider have such isomorphisms defined. We also note that Pooley (2006, p. 103) makes a closely related observation, and also cites Kaplan as recognizing that even when these isomorphisms are introduced, the sets we use to build models “will be open to a haecceitistic misinterpretation”.

inal theory, with its criterion for identity of models (viz. isomorphism), with a new theory that recognizes more properties of models and fewer isomorphisms between them. For example, the Special Theory of Relativity says that spacetime is a four-dimensional affine space with a Lorentzian metric. It does not recognize any difference between two such affine spaces, even if there are set-theoretic differences — e.g. the first contains $\{\emptyset\}$ and the second does not. Now, one could adopt a new theory in which each distinct set of cardinality $2^{\aleph_0}$ represents a physically inequivalent possible world. This new theory is indeterministic because there are two spacetimes M and M' that are identical up to $t = 0$, but where $\{\emptyset\} \in M$ while $\{\emptyset\} \notin M'$. We will not argue that this new theory is incoherent, but we insist that it is not the Special Theory of Relativity.

We’ve been at pains to argue that the distinction between “qualitative” and “de re” equivalence is too opaque to be of much use in deciding whether a theory is deterministic. That being said, we do agree with Hawthorne, Pooley, Teitel, and others that Lewis’ criterion for determinism, combined with “isomorphism is equality”, fails to detect some failures of determinism. Where we differ is that on our view, the problem is not that we need a binary relation that is finer-grained than isomorphism. Instead, it is that we need to keep track of the functions, i.e. the isomorphisms, that ground the claim that two models are isomorphic. While previous discussions have analyzed determinism in terms of binary relations between worlds, a more illuminating analysis can be carried out in terms of functions between worlds — what Butterfield and Belot call “duplications”.

5 Lewis formalized

Distinguishing senses of determinism by distinguishing types of descriptions or equivalence classes of models did not get us very far. But in fact, we think something very much in the spirit of the distinction that Hawthorne, Teitel, and others have aimed at is available, and that an adequate definition of determinism that captures something like the intuition behind Full Determinism, properly understood, is available. To see it, though, requires us to shift back to the other post-Lewisian thread, the one that was prematurely cut off by Belot (1995a) and Belot (1995b). We intend to take this line of development back up, and to argue that it is a genuine problem-solver for questions about determinism. In other words, insofar as Lewis was continuing

the Carnapian program of explication, then he was on the right track.

This line of development consists of three positive papers, and two negative papers. On the positive side, Butterfield (1987; 1988; 1989) points out that there is an imprecision in Lewis’ talk of “diverging worlds”. Since Lewis’ counterpart theory entails that no individual can exist in two worlds, distinct worlds can never really have overlapping initial segments. Butterfield then proposes that Lewis needs a notion of a *duplication map* $g : U \rightarrow U'$, where U is an initial segment of W , and U' is an initial segment of W' . With this notion in hand, Butterfield explains that there are two precisifications of Lewis’ notion of determinism, a stronger one “DM2”, and a weaker one “DM1”. Roughly speaking, DM2 says that if there is a duplication $g : U \rightarrow U'$ of initial segments of worlds W, W' , then there is a duplication $f : W \rightarrow W'$ of these worlds.¹⁴ Then Butterfield goes on to apply this analysis to the hole argument, and shows that GR satisfies DM2.

This is an amazing outcome. Butterfield seems to have blocked the hole argument, vindicating substantivalism and Lewisian counterpart theory, not to speak of formal definitions of determinism. But then along came Belot (1995b), who showed that even Butterfield’s stronger condition DM2 was too liberal by giving an example of a clearly indeterministic process that satisfies it. He then went on to give two refined and strengthened versions of DM2, but immediately provided counterexamples to them. The upshot seems clear: do not try to turn Lewis’ metaphysical definition of determinism into a Carnapian explication, because formal definitions will never capture the full sense of determinism. At least that seems to have been the lesson that many philosophers — Belot included — took away from his arguments.

We have already argued against this general posture. But we also think Belot’s arguments fail on their own terms. That is, we think that Belot himself gave a promising formal definition of determinism. We will now argue that his “counterexample” is nothing of the sort. It does *not* show the inadequacy of his precisification of Lewis’ diverging-worlds definition of determinism.

Belot’s (1995b) first definition of determinism is essentially a direct transcription of Butterfield’s DM2:

¹⁴This discussion occurred in the context of the hole argument, and so the definitions were originally stated in terms of manifolds, tensors, and smooth mappings. But the structure of Butterfield’s definitions are independent of these details. For further development in those terms, specifically for GR, see Manchak, Barrett, et al., 2025.

D1: A world W is deterministic if, whenever W' is physically possible with respect to W and t, t' , and $f : W_t \rightarrow W_{t'}$ are such that f is a duplication, there is some duplication $g : W \rightarrow W'$.

Belot argues that D1 does not capture determinism in its fullest sense, since there are indeterministic processes that are D1-deterministic. Belot presents an example of a centrally loaded column that must buckle in some direction — but which direction is undetermined. Since any two final states are related by a duplication map, this example satisfies D1 since. However, the example still seems to be indeterministic in some sense.

After dismissing D1, Belot considers the following strengthened criterion:

D2: W is deterministic if, whenever W' is physically possible with respect to W , and t, t' , and $f : W_t \rightarrow W'_{t'}$ are such that f is a duplication, there is some duplication $g : W \rightarrow W'$ whose restriction to W_t is f .

The key difference between D1 and D2 is that the latter requires a relationship between the duplication $g : W \rightarrow W'$ and the duplication $f : W_t \rightarrow W'_{t'}$, viz. f is the restriction of g to W_t . The fact that D2 is genuinely stronger than D1 depends on the assumption that “agreement” can be witnessed by various functions. Indeed, if agreement were a binary relation on worlds (or world segments), then D1 would imply D2. This might explain why modal metaphysicians have overlooked D2 (or the even stronger version D3, that we will soon consider). Modal metaphysicians have tended to think in terms of binary relations on worlds, whereas D2 asks us to keep track of different ways that worlds can be matched with each other.

Belot then provides a counterexample to D2: a single α particle decays into two identical β particles. This example satisfies D2: if f is a duplication of initial segments, then f can be extended to a duplication of worlds. In fact, since the β particles are assumed to be identical (i.e. related by a symmetry), f can be extended in two ways. But this very non-uniqueness of extensions makes Belot (and us) think that the process is not actually deterministic. If β_1 and β_2 are the particles in one world, then we share Belot’s intuition that the following counterfactual is true:

(*) β_1 could have been β_2 .

(Note, however, that the truth of (*) does not depend on the existence of a distinct world.)

We agree, then, that D2 does not capture determinism in the strongest sense. But Belot has yet another proposal.¹⁵

D3: A world W is deterministic if, whenever W' is physically possible with respect to W , and t, t', W' and $f : W_t \rightarrow W'_{t'}$ are such that f is duplication, then there is exactly one duplication $g : W \rightarrow W'$ which extends f .

That is, D3 strengthens D2 by requiring uniqueness of extension of a duplication map of initial segments.

Some of the details of D3 are inessential, and it can easily be made into a schematic that applies to just about any scientific theory. For example, while D3 is formulated in terms of possible worlds, we will sometimes talk instead about models (of a theory). Similarly, D3 takes the determiner to be a time-slice W_t , but we can take it to be other parts of a world or a model, e.g. an initial segment of a possible world (see Lewis, 1983; Butterfield, 1989), or an initial data surface embedded in a four-dimensional manifold (see Landsman, 2023). The details may differ, but all of these cases conform to the following schematic:¹⁶

$$\begin{array}{ccc} M & \overset{g}{\dashrightarrow} & M' \\ \uparrow i & & \uparrow i' \\ U & \xrightarrow{f} & U' \end{array}$$

Here $i : U \rightarrow M$ and $i' : U' \rightarrow M'$ are the embeddings of initial segments into the entire history, and $f : U \rightarrow U'$ is an isomorphism of initial segments. D3 then says: determinism holds just in case any isomorphism of initial segments extends uniquely to an isomorphism of worlds.

In what follows, we will argue that D3 captures the fullest sense of determinism that is reasonable to apply to a physical theory. For now, we will simply point out that D3 holds for General Relativity. To be more precise, D3 holds for four-dimensional, maximal, globally hyperbolic, vacuum solutions to Einstein's equation. This result follows from the conjunction of the Choquet-Bruhat-Geroch theorem and Geroch's (1969) rigidity theorem — see the accounts in (Landsman, 2023) and (Read and Manchak, 2025).¹⁷

¹⁵Belot's D3 seems to have been independently rediscovered by Landsman (2023) and Cudek (2024).

¹⁶As we note later, this schematic looks suggestively like a version of implicit definability.

¹⁷A more careful treatment of determinism in GR is given in Manchak, Barrett, et al.,

6 Belot against D3

We think that D3 is an excellent definition of when a theory is deterministic. But that is not the conclusion that its architect, Belot, drew. In fact, Belot’s criticism of D3 made it all but invisible to philosophers for thirty years, until it reappeared in work by Halvorson and Manchak (2022), Landsman (2023), and Cudek (2024). In this section, we consider Belot’s purported counterexample to D3, and we show that it underspecifies the relationship between spacetime and its material contents.

In this example, W is a world with spacetime points and Newtonian spacetime structure. It initially contains a single α particle. The laws of nature decree that one year later, at $t = 1$, the α particle decays into continuum many β particles; arranged so that at time t , the β particles form a spherical shell of radius t ; with each β particle moving away from the center of the sphere along its radius.¹⁸ (Belot, 1995b, p 193)

Let’s follow Belot’s own presentation of the conflicting claims that this world is deterministic according to D3, but is actually indeterministic.

Belot’s argument that W satisfies D3 relies on the following claims:

1. Duplications must preserve metric relations between spacetime points. Hence, a duplication map from W to itself must be a symmetry of Newtonian spacetime.
2. Duplications must preserve the relation of “ x is located at y ” that holds between a material object and a spacetime point.
3. The only Newtonian symmetry that preserves the worldline of the α particle is a rotation around the timelike line that extends that particle’s trajectory.

Supposing that these three claims hold, then if f is a duplication of W ’s initial segment, then f is a rotation. It follows that f has a unique extension

2025. Our attitude here should be clarified by what we say in section 3 about electromagnetism. The difference between two versions of EM manifests itself in different choices of isomorphisms between models; and one of these two choices leads to a better theory (in our opinion, and in the opinion of most physicists).

¹⁸We changed the time scale for ease of exposition.

to all of the spacetime points of W ; and since location relations must be preserved, f has a unique extension also to material objects. Therefore the extension of f is unique, and D3 is satisfied.

Belot then argues that the example should be conceived of as indeterministic. His argument relies on the following claim:

There is a legitimate counterpart relation g' (not a duplication) that is the identity on spacetime points, but not the identity on material objects. That is, g' does not preserve the relation “ x is located at y .”¹⁹

We agree completely that if g' is “a legitimate counterpart relation”, then the correct judgment is that the example is indeterministic. In fact, the existence of two counterpart relations g and g' , both of which extend f , would indicate a failure of D3. The problem is not with D3 as a criterion, but with an ambiguity about which functions from W to itself are relevant for deciding whether W is deterministic. That is, whether or not a world is deterministic depends crucially on which functions from that world to itself are actually duplications. Nor do we think that this question — what are duplications? — can be settled by apriori metaphysics. Formulating a scientific theory includes a specification of which maps between models are isomorphisms (i.e. duplications), and that specification is decisive for whether that theory is deterministic.

We have seen that there are two ways to read Belot’s example: either the relation between β particles and spacetime points is nomologically necessary, or it is not. If this relation is nomologically necessary, then the symmetries of a spacetime are uniquely paired with symmetries of its material contents. If this relation is not nomologically necessary, then there could be a spacetime symmetry that is not uniquely paired with a symmetry of its material contents. A theory that describes the former situation can be expected to satisfy D3, while a theory that describes the latter situation can be expected not to satisfy D3. In this section, we will give simple examples of such theories.

Newtonian spacetime has the feature that spatial points maintain their identity over time, and so it makes sense to talk about whether an object is changing its position over time. (This in contrast to Galilean spacetime.) Newtonian spacetime also has a rather small group of symmetries: uniform

¹⁹“ g' is a counterpart relation that tells us that β_1 could have been β_2 etc. g and g' license us to say that there are two possible futures for W_t .”

shifts and rotations around timelike (vertical) lines. These features of Newtonian spacetime play an important role in the setup of Belot’s example, but we can capture the essential points with a simple toy model. It will suffice to have three locations (left, center, and right), and two times ($t = 0$ and a final time $t = 1$). We won’t need shifts (which play no role in Belot’s example), but we allow for rotations around the center location, i.e. permutations of right and left.

There are two ways to set up such a framework, which are equivalent for Newtonian spacetime, and only come apart for more sophisticated examples.

1. Represent spacetime by a family of types S_t , with t a time parameter, and postulate a “persistence” relation (the analogue of an affine connection) between the types.²⁰ The persistence relation can be represented by isomorphisms with compatibility relations. To represent Newtonian spacetime, we assume a unique isomorphism $\delta_{t,t'} : S_t \rightarrow S_{t'}$.
2. Represent space by a type S and time by another type R , so that spacetime is represented by the product type $R \times S$.

The advantage of the first, more complicated, setup is that it generalizes more easily to other spacetime theories, e.g. Galilean spacetime. The advantage of the second setup is that we don’t have to keep track of sorts. Since Belot’s example assumes Newtonian spacetime, we will start with the second approach.

Let Σ be a signature with a sort symbol S for spatial points. We could then add the axiom that there are three things of sort S , corresponding to the three particle positions. But it is simpler just to ignore that α particle, which effectively defines a constant symbol (a name for its location). Thus we take as our first axiom:

There are two things of sort S .

We now let B be a sort symbol for the β particles, and we add a second axiom:

There are two things of sort B .

²⁰Here we use “type” and “sort” synonymously, both in the sense of many-sorted logic (see Halvorson, 2019). We are using this framework for its flexibility, and without any commitment to type theory as a foundation of mathematics.

β_1		β_2
	α	

Figure 1: Space with three places and two times, with two β particles at the later time. Since symmetries are assumed to fix α , the center blocks can be omitted from the model without changing the conclusions we draw.

The crucial decision now is whether to add an additional relation symbol that relates β particles to spatial points. To do so would amount to positing a lawlike relation between material objects and spatial points — a relation that must be preserved by all symmetries of models and all isomorphisms between models.

Suppose first that we do *not* add any such relation, and let T be the bare theory of spatial points and β particles. It is clear that T does not satisfy D3: any symmetry g of spatial points is compatible with two distinct symmetries of β particles (the identity and the flip).

Suppose now that L is a relation between β particles and spacetime points, and let T_q be the theory that says that each β particle is located at a unique spatial point at $t = 1$. A model M of T_q consists of two two-element sets S^M and B^M , and a one-to-one correspondence L^M between B^M and S^M . It is easy to see that the models of T_q satisfy D3. Indeed, let $i : S \rightarrow M$ and $i' : S' \rightarrow M'$ be embeddings of initial conditions. Here S and S' are sets with two elements, and M (respectively M') adds a second set B (respectively B') and an isomorphism $L : B \rightarrow S$ (respectively $L' : B' \rightarrow S'$). Any isomorphism $f : S \rightarrow S'$ of space is compatible with precisely one isomorphism $L' \circ f \circ L$ of β particles:

$$\begin{array}{ccc}
 B & \dashrightarrow & B' \\
 \uparrow L & & \uparrow L' \\
 S & \xrightarrow{f} & S'
 \end{array}$$

Therefore, T_q satisfies D3.

The theories T and T_q illustrate the obvious fact that whether or not a world W is deterministic depends on the relations that world bears to other worlds. If all nomologically possible worlds have the same pairing of β particles and spacetime points, then the process of beta decay is deterministic. If some nomologically possible worlds have different pairings of β particles and spacetime points, then the process of beta decay is indeterministic. This difference in verdicts is not a bug in definition D3; it is a feature of D3 that it depends on a precise specification of permitted duplications.

There is yet another theory that could be said to capture Belot's example. Let T_h be a theory with a sort symbol S for spatial points, and with a name b for one of the two β particles. We consider $\ulcorner x = b \urcorner$ to represent the property that the spatial point x is occupied by b at $t = 1$. In this case, a model M of T_h consists of a set S^M with two elements, and a distinguished element $b^M \in S^M$. If M and M' are models of T_h , then an isomorphism $g : M \rightarrow M'$ consists of a function from S to S' such that $g(b^M) = b^{M'}$. It follows that there is a unique isomorphism between any two models of T_h .

For T_h , an initial condition is a set S with two elements, while a final condition is the same set S plus the choice of one of the two elements $b^M \in S$. This choice leads to a reduction of symmetry, and so to a violation of D3. Indeed, consider the isomorphism $f = 1_S : S \rightarrow S$ of initial conditions. Let $b^M \in S$, and let $b^{M'}$ be the other element of S . Then there is no isomorphism $g : M \rightarrow M'$ that completes the following diagram:

$$\begin{array}{ccc} M & \overset{g}{\dashrightarrow} & M' \\ \uparrow i & & \uparrow i' \\ S & \xrightarrow{f} & S \end{array}$$

Therefore, T_h does not satisfy D3.

Let's take stock. The theories T, T_q and T_h describe similar worlds: at the initial time there are two locations and no material particles; and at a subsequent time, each location is occupied by a distinct β particle. But there is a subtle difference in how these theories describe the world: T_q fixes the relationship between β particles and spatial points, and its symmetries preserve this relationship. In contrast, T and T_h permit different matchings between β particles and spatial points, and the symmetries of spatial points are not uniquely paired with symmetries of β particles.

Belot's clever example underspecifies the relationship between material particles and spatial points. According to one specification, material particles

are nomologically tied to particular spatial points. In this case, the world W is unambiguously deterministic. According to another specification, one and the same material particle can be located at different spatial points; and, in this case, a symmetry of space does not extend uniquely to a symmetry of its material contents. In this latter case, the world W should be judged to be indeterministic. The lesson is that determinism depends on which relations are preserved by duplications, and theories tell us what those relations are.

We conclude this section on an ironical note: Belot argues that “determinism is not a formal property of uninterpreted theories.” But the example he describes cannot be judged either to be deterministic or indeterministic until we are told precisely which mappings between worlds are duplications, and this latter specification is a formal property of a theory.

7 Bridging a non-existing gap

Teitel (2019) argues that metaphysicians have an important job in uncovering what modal-metaphysical commitments might be required to maintain the consistency of spacetime substantivalism with full determinism. He poses the challenge as “bridging the gap” between qualitative and full determinism.²¹

We need a doctrine that . . . bridges the crucial gap between GR’s qualitative determinism and its full determinism (thereby resolving both the original hole argument and my revised hole argument). (Teitel, 2019, p 379)

Any of those three anti-haecceitistic doctrines suffices to bridge the gap between GR’s qualitative and full determinism. (Teitel, 2019, pp 359-360)

²¹In more recent work, Teitel (2022) has backed away from this challenge. Instead, he bites the bullet and accepts that the haecceitist substantivalism that he prefers is indeterministic, properly construed. But he also maintains that any viable metaphysic that can make sense of the modal requirements imposed by “sophisticated” substantivalists suffers from its own challenges related to determinism, viz., that they make determinism too “cheap” by denying the metaphysical possibility that given spacetime points could have had different properties than they do. There is much to say in response to this rich paper, but for present purposes his earlier work is a better foil to make our point—even if he no longer defends the views we cite.

I deliberately set up the issues surrounding the hole argument by directly discussing modality and which doctrines imply the right modal correlations to bridge GR’s qualitative and full determinism, rather than following the standard practice in the literature of theorizing primarily in terms of mathematical solution spaces and what we use them to represent. (Teitel, 2019, p 388)

We agree that if there were a gap between the sense in which GR is deterministic and some more metaphysically significant kind of determinism, then it would be worth inquiring into what metaphysical commitments are needed to bridge this gap. We claim, however, that there is no such gap.²²

The hole argument raises many technical issues that are beyond the scope of this paper. Fortunately, the literature of the past thirty years offers numerous toy examples that are supposed to be analogous to GR in being qualitatively, but not fully, deterministic. (See Figure 7). We have encountered two already: Belot’s β particles and Hawthorne’s ships. The doubly-symmetric world described by Melia (1999) provides yet another. We will now show how Hawthorne and Melia’s examples, like Belot’s, illustrate the distinction between D1 and D3.

As a warmup, consider one of Melia’s simpler (but more entertaining) examples:

We could imagine a collection of bald philosophers, sitting in a circle. It is a law that one of them will grow a single hair. But, by the symmetry of the situation, *any* of the philosophers could be the lucky one. Again, our intuition is that there are many qualitatively isomorphic but distinct possibilities, each representing a different way in which the situation could evolve. (Melia, 1999, p 650)

Intuitively this example is qualitatively deterministic, since any two possible final conditions are qualitatively identical. And yet, this example clearly isn’t fully deterministic, since the law does not stipulate *which* philosopher will grow a hair. We agree that this example is deterministic in one sense, but

²²Of course, this does not mean we do not recognize different senses, or “strengths”, of determinism. For instance, Belot’s D1 and D2 are weaker than our preferred D3, and may well be viewed as capturing senses of “qualitative determinism”. The crucial point is that GR is deterministic in a stronger sense than either of these, and so there is no gap for determinism in GR.

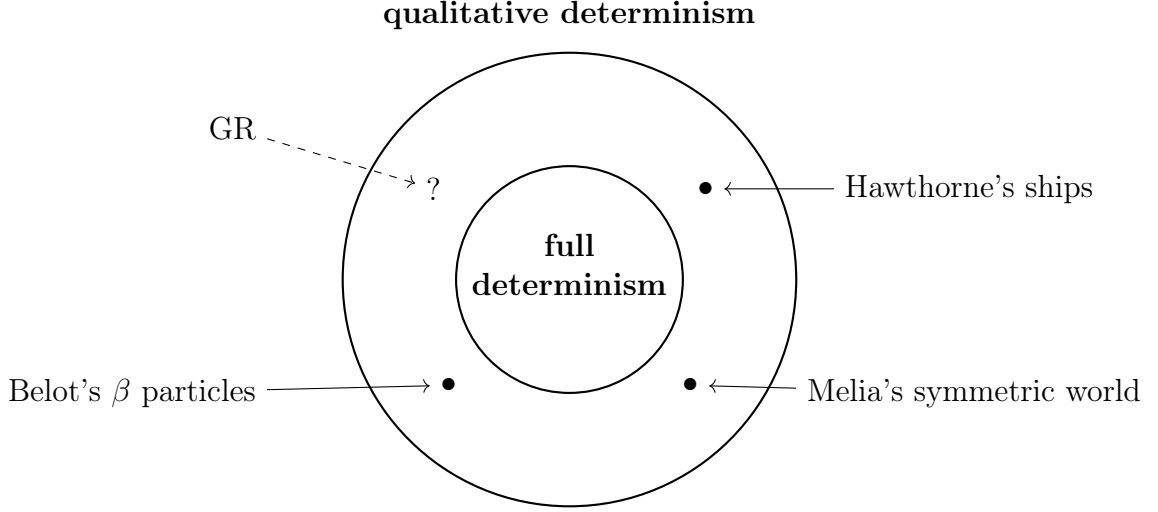


Figure 2: Theories that are supposed to be qualitatively, but not fully, deterministic.

not another. Only we think that the correct analysis is that this example is D1 but not-D3.

To see this, suppose that there are initially $n > 1$ philosophers, and that p_t represents the property of being bald at time t . Let T be the theory with axioms $\forall x p_0(x)$ and $\exists! x \neg p_1(x)$. Since T entails that p_0 holds of all things, the predicate symbol p_0 plays no role in the analysis, and we may drop it. If we set $p = p_1$ for notational simplicity, then a model M of T is determined by a set S and a singleton subset $p^M \subseteq S$. If M, M' are models of T with the same initial conditions (i.e. the same domain S), then there is at least one bijection $g : S \rightarrow S'$ such that $g(p^M) = p^{M'}$. Thus, $g : M \rightarrow M'$ is an isomorphism, and D1 is automatically satisfied.

We now show that D3 fails. Let M be a model of T whose domain S has two elements; and let M' be the model that has the same domain as M , but where the extension p^M has been switched to the other element of the domain, i.e. $p^M \neq p^{M'}$. (Note that M and M' are isomorphic models.) Then the identity 1_S is an isomorphism between the initial conditions of M and M' . However, if 1_S were an isomorphism of M to M' , then it would follow that $p^M = p^{M'}$, contradicting the definition of M' . Therefore, T is D3-indeterministic.

Melia's bald philosophers example is supposed to be a paradigm case where qualitative, but not full, determinism holds. But our split intuitions about this example can be explained by a more clear distinction, viz. that between D1 and D3. The bald philosophers example does not provide any support for the legend that there is a deeper, metaphysical sense of determinism that cannot be captured by a formal definition.

But Melia has another trick up his sleeve: an example so clever in conception that one feels sure that the quest for a formal definition of determinism will have to be abandoned.

Consider a world whose initial conditions consist of the following situation (see Figure 3). The two white particles are duplicates of each other and the two black particles are duplicates of each other. The laws in this world dictate that, after a certain fixed period of time, each black particle will start moving at a fixed velocity in a straight line towards a white particle, and that the two black particles will move towards *different* white particles. Using names for the objects found in the situation above, after a fixed amount of time either **c** will head towards **b** and **d** will head towards **a**, or **c** will head towards **a** and **d** will head towards **b**. (Melia, 1999, p 661)

Once again, it is clear that the two possible final conditions are qualitatively identical, and hence that this example is qualitatively deterministic. But surely, one thinks, there is a *haecceitistic* difference between the two final conditions, and so the example is not fully deterministic.

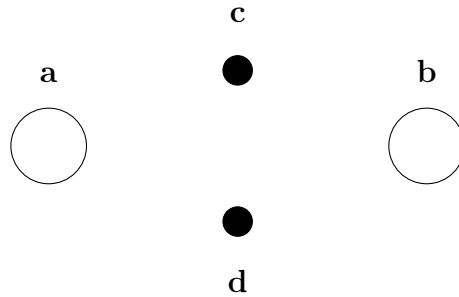


Figure 3: Melia's symmetric world

We will show, once again, that what this example illustrates is the distinction between the definitions D1 and D3. Here we just have to be a bit

more careful with how we explicate the details of the example.

The most straightforward regimentation of Melia’s example has two domains W and B for the particles, and for each $t = 0, 1$, a relation $\alpha_t(x, y)$ to indicate that at time t , x is adjacent to y . We add the axioms that at $t = 0$, no two things are adjacent, and that at $t = 1$, the adjacency relation induces a bijection between W and B .²³ Since T dictates that α_0 is empty, its presence is structurally irrelevant. Thus we drop α_0 and set $\alpha = \alpha_1$. Let’s call the resulting theory T .

A model M of T consists of two sets, each with two elements, and a bijection α^M between them. Here the initial conditions are just the two sets, whereas the final conditions include the bijection α^M . It is clear that any two models M, M' of T are isomorphic, and so T automatically satisfies D1.

It will now be easy to see that T does *not* satisfy D3, and the reason is that the final conditions have less symmetry than the initial conditions. More rigorously, in a model M of T , the initial conditions (i.e. the two sets W and B) are invariant under any symmetry of the form $\langle f_0, f_1 \rangle$, where $f_0 : W \rightarrow W$ and $f_1 : B \rightarrow B$ are bijections. In contrast, the final conditions include a bijection $\alpha^M : W \rightarrow B$, and this bijection is not invariant under symmetries of the form $\langle f_0, f_1 \rangle$ where f_0 and f_1 do not have the same polarity (i.e. where f_0 is the identity and f_1 flips elements or vice versa). Therefore, there is a symmetry of initial conditions that does not extend to a symmetry of models, and T is D3-indeterministic.

The upshot: to understand the sense in which Melia’s doubly-symmetric world is indeterministic, we do not need to know anything about haecceitistic differences. It is enough to see that there is a duplication of initial conditions that does not extend to a duplication of worlds. So this example, and others like it, only emphasizes the virtues of “purely formal” definitions of determinism.

Similar remarks can be made about Hawthorne’s ships. As with Belot’s and Melia’s examples, his example is D1 but not D3 deterministic — reinforcing our claim that this distinction suffices to capture our intuitions.

To be precise, let T be the theory with a single unary predicate p that says: there are exactly two things, and one of them is p . Here we take $p(x)$ to mean that x sinks at $t = 1$. This theory is as simple as can be imagined. A

²³We can imagine the white and black particles as sitting on the vertices of a kite, and we can think of α as a metric, where at $t = 0$, each white particle is distance $\sqrt{5}$ from each black particles; and at $t = 1$, each white particle is distance 0 from one black particle, and distance 4 from the other.

model M of T consists of a set with two elements and a singleton extension for p . For any two models M and M' of T , there is a unique isomorphism $f : M \rightarrow M'$. This shows that T is D1-deterministic.

But T is not D3 deterministic. To see this, let $S = S' = \{a, b\}$, let $i : S \rightarrow M$ be the embedding into a model M such that $M \models p(a)$, i.e., a sinks, and let $i' : S' \rightarrow M'$ be the embedding into a model M' such that $M' \models p(b)$, i.e., b sinks. Then $1_S : S \rightarrow S'$ is an isomorphism of initial conditions, but there is no $g : M \rightarrow M'$ that completes the following diagram:

$$\begin{array}{ccc} M & \overset{g}{\dashrightarrow} & M' \\ \uparrow i & & \uparrow i' \\ S & \xrightarrow{1_S} & S' \end{array}$$

Therefore, T is D3-indeterministic.²⁴

In summary, the examples by Belot, Melia, and Hawthorne have been thought to provide evidence for the existence of a gap between qualitative and full determinism. What these examples actually illustrate is the distinction between D1 and D3.

8 D2, D3 and Full Determinism

We have suggested that D1 captures the concept of qualitative determinism, and the literature agrees with us on this. Where intuitions might still clash is whether D2 or D3 captures the strongest sense of determinism that it makes sense to ask about. We claim that D2 is not yet full determinism, but D3 is.

A paradigm example that satisfies D2 but not D3 is Belot's: the growing sphere of β particles in Newtonian spacetime, at least when formalized as consisting of two kinds of things, β particles and spacetime points, that are *not* connected by a lawlike relation. This example satisfies D2, since any symmetry of empty space trivially extends to a symmetry of space and its material contents. This example does *not* satisfy D3, since symmetries of empty space do not determine symmetries of material contents. But what does the non-uniqueness of extensions of isomorphisms have to do with determinism? Lewis' account, read naively, says that a failure of determinism entails the

²⁴For what it is worth, Hawthorne's ships example is also indeterministic on D2, for the same reason. Thanks to an anonymous referee for suggesting we make this explicit.

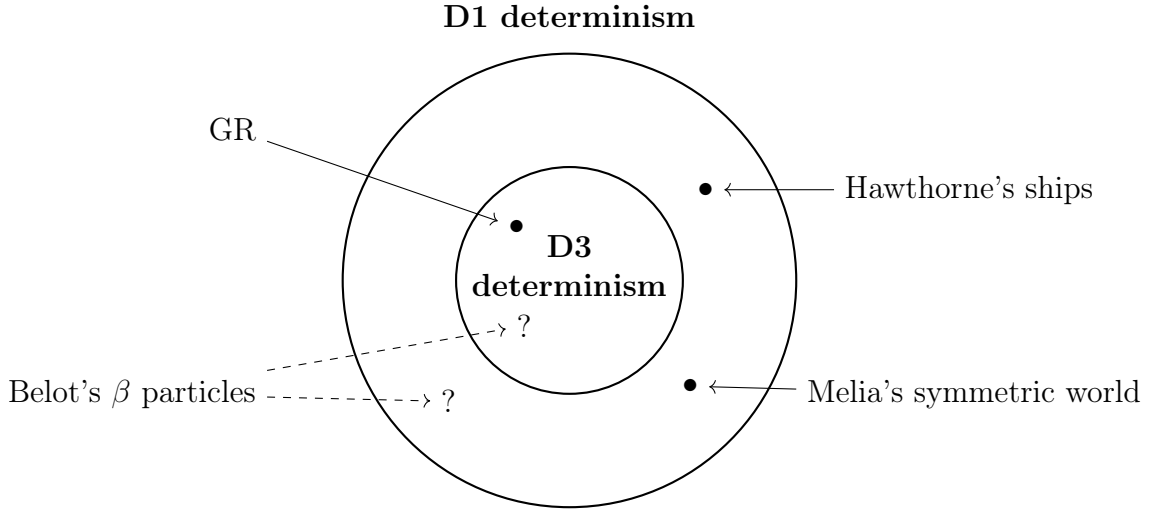


Figure 4: The toy examples in the literature are D3-indeterministic, while GR is D3-deterministic. Belot's example can be interpreted in two ways, one deterministic and one indeterministic.

existence of two distinct worlds. But here it is hard to see what entitles us to say that there are two distinct worlds. After all, any two models of this toy theory are isomorphic. The problem is not a failure of isomorphism, it is that there are multiple isomorphisms.

We should not allow ourselves to become confused by trying to count the number of possible worlds, or by trying, in general, to analyze determinism in terms of static concepts. Determinism is a claim about how things at one time are related to things at another time. As Carnap et al. realized, it is more clear to describe determinism as a property of a scientific theory: that theory posits a fixed relation between past states and future states. If we set aside the murky talk about identity of possible worlds, it should be clear from common sense that Belot's example can be described in two ways: either the relationship of the α particle to spatial points does determine the relationship of the β particles to spatial points, or it doesn't. In the latter case — made explicit by our toy theory with two sorts S and B — there is only one model (up to isomorphism), but in that model, a symmetry of an initial time slice does not determine a symmetry of a latter time slice. That's a clear sign that that facts about the latter time are not entailed by the laws

and facts about the earlier time.²⁵

The upshot of these considerations is that neither D1 nor D2 captures the strongest sense in which the past can determine the future. We argue, in contrast, that D3 does.²⁶

It is not so easy to compare D3 to Full Determinism, since the latter hasn't been given a precise, formal definition. When we tried to make Full Determinism precise, the only way we could do so was by positing a distinction between a de re language Σ_r and a qualitative sub-language Σ_q . So let's first look at the special case of theories that come equipped with a de re language, i.e. names for all objects. In that case, we can show that FD, D1, D2, and D3 are all equivalent. But this result also shows why it is important to look at theories *without* names, where FD does not apply, because only those theories illustrate the distinctions between D1, D2, and D3.

To make things more clear, we remind the reader of a simple point: a Σ_r -theory T is fully deterministic iff T satisfies D1.

Proposition 1. *For theories with sufficiently many names, FD, D1, D2, and D3 are equivalent.*

Proof. As noted above, the definition of FD is formulated against a background assumption that there are two signatures $\Sigma_q \subset \Sigma_r$. However, FD is just D1 for a Σ_r -theory, and we are assuming that T is such a theory. It will suffice then to show that D1 implies D3. Suppose then that T satisfies D1, and let $f : U \rightarrow U'$ be an isomorphism. By D1, there is an isomorphism $g : M \rightarrow M'$. Since all elements of U and U' are named, the isomorphism $f : U \rightarrow U'$ is unique, hence $g|_U = f$. That is, g extends f . Similarly, since all elements of M and M' are named, the isomorphism $g : M \rightarrow M'$ is unique. Therefore g is the unique extension of f , and T satisfies D3. $\square$

But what about the case where we do not have names? One natural suggestion for extending FD to such theories is simply to add names when necessary. Given a Σ -theory T , let Σ^+ be the expansion of Σ to include

²⁵We have tacitly moved here from a claim about models to a claim about entailments. We suspect that this move can be formalized as a general version of Beth's theorem, showing the implicit definability entails explicit definability. In this case, the semantic formulation of determinism should entail a corresponding syntactic formulation.

²⁶In fact, at least for spacetime theories, more can be said about the difference between D2 and D3. Read and Manchak (2025) show that D3 holds if and only if: D2 holds along with any one of three equivalent rigidity statements which are articulated in: Geroch (1969), Weatherall (2018), and Halvorson and Manchak (2022).

“sufficiently many” names, and let T^+ be the extension of T to Σ^+ . We can then ask about the relation between T having property D2 or D3 and T^+ having property FD (i.e., D1). In some cases, D2 (and D3) for T do imply FD for the enriched theory. In particular, this happens for theories of enduring objects – that is, theories whose models’ domains coincide with the domains of their initial segments. D2 for a theory T implies that the name-enriched theory T^+ satisfies FD.

Proposition 2. *Suppose that T is a theory of enduring objects. If T satisfies D2, then T^+ satisfies FD.*

Proof. Suppose that T satisfies D2. Let M, M' be models of T^+ with initial segments U, U' such that $U \sim_r U'$. That is, there is a Σ^+ -isomorphism $f : U \rightarrow U'$. Since T and T^+ are theories of enduring objects, $|M| = |U|$ and $|M'| = |U'|$. It follows that $f(c^M) = c^{M'}$, for all names c . Since f is a Σ -isomorphism, D2 entails that there is a Σ -isomorphism $g : M|_\Sigma \rightarrow M'|_\Sigma$ that extends f . Since g extends f , $g(c^M) = f(c^M) = c^{M'}$, for all names c . Therefore, g is a Σ^+ -isomorphism, and $M \sim_r M'$. Therefore, T^+ satisfies FD. $\square$

What drives this result is that if objects persists over time, then fixing the referents of constant symbols in initial segments fixes those referents in the entire model. But as we have seen, not all theories have enduring objects. In general, we find that even D3 for a theory T comes apart from FD for an enriched theory T^+ . First we now show that T^+ satisfying FD does not imply that T satisfies D3.

Example (toy Leibnizian spacetime). Let T be a theory with two sort symbols S_0, S_1 , and axioms that say that there are exactly two objects of each sort. The theory T satisfies D2, but not D3, since an automorphism of the first sort S_0 extends in more than one way to an automorphism of the second sort S_1 . The theory T^+ adds two constant symbols of each sort, say a_0, b_0 and a_1, b_1 . But then for any models M, M' of T^+ , there is an isomorphism $f : M \rightarrow M'$, and so $M \sim_r M'$. Therefore, T^+ satisfies FD. $\square$

The converse also fails: that is, a theory T may satisfy D3, but the enriched theory T^+ with sufficiently many names might not satisfy FD.

Example (toy Newtonian spacetime). We now consider Newtonian spacetime as having a different domain S_t of spatial points for each time, and

isomorphisms $\delta_{t,t'} : S_t \rightarrow S_{t'}$ that pick out the preferred frame of reference. For our purposes, it suffices to consider a simple case with two sorts S_0, S_1 , and a single function $\delta : S_0 \rightarrow S_1$. Let T be the theory that says there are exactly two elements of type S_0 , and that δ is a bijection. If M, M' are models of T , then an isomorphism $g : M \rightarrow M'$ consists of two bijections $g_0 : S_0 \rightarrow S'_0$ and $g_1 : S_1 \rightarrow S'_1$ that satisfy the compatibility condition $\delta^{M'} \circ g_0 = g_1 \circ \delta^M$. It follows that any bijection $g_0 : S_0 \rightarrow S'_0$ extends uniquely to a bijection $g : M \rightarrow M'$. Therefore, T satisfies D3.

For the name-enriched theory T^+ , suppose that a_0, b_0 are constant symbols of sort S_0 , and that a_1, b_1 are constant symbols of sort S_1 . Then there is one model M of T^+ such that $M \models \delta(a_0) = a_1$, and a non-isomorphic model M' of T^+ such that $M' \models \delta(a_0) = b_1$. But the initial segments of M and M' are isomorphic. Therefore, T^+ does not satisfy FD. $\square$

Note that in the toy Newtonian example, T^+ does not satisfy D3 (in addition to not being fully deterministic). Therefore, adding names to a D3 deterministic theory can result in a D3 indeterministic theory. It might seem like a strike against D3 that it is not stable under the addition of names to a theory. But that intuition is based on a false assumption that the role of names in formal theories is the same as the role of names in ordinary language. In a formal theory, introducing a new name is tantamount to introducing a new property $\varphi(x) \equiv (x = c)$. But we should expect that adding new properties, without adding dynamical laws that govern the behavior of those properties, could transform a deterministic theory into an indeterministic theory.

Something similar happens with GR. As usually formulated, GR satisfies D3 (Halvorson and Manchak, 2022). But adding names to GR results in a theory that does not satisfy D3, and thus is not fully deterministic — as shown by the hole argument. Indeed, Weatherall (2018) suggests that one way of understanding manifold substantivalism, as described by Earman and Norton (1987), is as a view on which there are additional singular, haecceitistic facts about spacetime points that can only be described by something like enriched — or, in Pooley’s terms, substantivalist — GR. The hole argument then shows this theory is indeterministic, even on D1. Of course, the important point is that this enriched theory is not GR, the theory that we have good reasons to believe at least approximately describes the structure of space and time in our universe, but rather GR plus a great deal more structure. Without that structure, GR is deterministic by D3, and FD does

not even apply.

9 Interpretation revisited

Some readers may find the analysis at the end of the previous section unsatisfactory. One might argue, for instance, that we have simply misunderstood full determinism. What motivates full determinism is the idea that there are individuals in the world, and haecceitistic facts about those individuals. A deterministic theory ought to be able to assign them properties in an unambiguous (deterministic) way. The names just give us a way of referring to those individuals. When we say that FD simply does not apply to GR, that is not a problem for FD, it is a problem for GR! We need enriched GR to accurately assess whether the theory determines the properties of individuals. Without names, we are simply dwelling in the domain of the qualitative.

We think this posture is wrongheaded. But rather than argue against it directly, we want to propose a diagnosis of where it originates, drawing on the arguments from section 3. As we argued there, interpretation is itself formal. All anyone is doing when they try to interpret theories is just layering models on models. Crucially, for the present point: Tarskian semantics involves mapping theories, with or without names, into set theory, typically understood as a theory with names (or rather, as theory whose membership properties allow us to uniquely individuate sets).

We suggest that the motivation for FD arises because when you interpret a theory without names in set theory, without paying careful attention to how the semantics works, it looks as if the “real” points, the ones the theory is referring to, have names. This apparently means we can ask about what “determines” what properties those named things have. But this instinct is a mistake. It illegitimately mixes two different things: the theory we are trying to analyze, and the formal tools we use to analyze it. Determinism for theories is about whether initial segments of models determine the entire model. The “determination” of which objects in a model carry which properties (or names) is about an interpretation map, in the Tarskian sense. In other words, failures of FD are about *us*, that is, about how we think about our formal semantics and how we define interpretation maps, not about our theories or the world.

We suggest that something similar happens in many discussions of the hole argument. Philosophers apparently mistake GR for enriched GR when

thinking about substantivalism. Doing this is not only a mistake, it quickly leads to incoherence. We can see this point most starkly by considering a special sector of GR, consisting of models that satisfy a strong asymmetry property called “Heraclitus” (Manchak and Barrett, 2023; Manchak, Barrett, et al., 2025). Spacetimes with that property are such that every point is uniquely specified by its metrical properties (including derivatives, i.e., curvature scalars). Call the theory of Heraclitus spacetimes HGR. HGR has names, in the sense that one can uniquely refer to points. This theory satisfies FD. (Of course, it also satisfies D3.)

Now consider what happens when we interpret HGR in set theory. We assign those named points to sets, which also have names. But of course, nothing in the theory can determine which (named) set we assign to which named point. No theory can do that, because it happens at the level of choosing an interpretation map! What this means, though, is that on the set theory side we now have *too many names*. That is, if we try to doubly-interpret this theory, as we suggest the FD advocate would wish to, and run our analysis of determinism on those doubly-interpreted structures, we will find that they are not deterministic. We claim this is a completely generic situation that arises from layering interpretation on top of set theory.

And it gets worse! Suppose we somehow solved this problem in the doubly-interpreted theory, perhaps by adding laws that coordinate between the two types of names, restoring FD. What then? Now we have a new theory, with lots of redundant names, that has no expressive resources beyond our original theory. (Perhaps the theories are even logically equivalent, depending on the details.) But then we can interpret that theory, using Tarskian semantics. The (triply) interpreted theory will now have three types of names — the two coordinated ones in our theory, plus the names of the sets on the semantic side. The problem will arise again. And so on *ad infinitum*. But once we see how this works, it is clearly a pseudo-problem, one arising only because of a confusion about what is part of the theory and what is not, what the theory *should* determine and what is merely structure added in interpretation.

10 Conclusion

Analytic philosophy was, in one sense, born from the idea that philosophers should avail themselves of the tool-kit of mathematics. We find it odd, then,

that prominent analytic philosophers have recently argued *against* formal approaches, saying things like, “the purely formal approach is a nonstarter”, or, “determinism cannot be a formal property of theories.” The descriptive content of such claims is opaque (what is a non-formal property?), but their illocutionary force is to recommend against adopting the methods that distinguished analytic philosophy from the more speculative, and less science-friendly, approaches of the nineteenth century. Surely this belongs among the ironies of intellectual history.

We are also motivated by a practical concern about how to facilitate fruitful dialogue between philosophy and the natural sciences. If philosophers insist on making distinctions that cannot gain any traction in scientific practice, then they will only reinforce disciplinary boundaries that are harmful to both philosophy and the sciences.

To be clear, we are *not* arguing for a kind of science-deference that says, “if scientists don’t regularly make some distinction, then neither should philosophers”. For one, we recognize that scientists might have practical reasons to blur over distinctions of genuine metaphysical significance. One might have thought that the distinction between qualitative and full determinism is of this sort; but our investigation has shown that the relevant conceptual joint is not here. The relevant conceptual joints are drawn with the help of isomorphisms between models, and these joints clearly distinguish which real-life scientific theories are deterministic. With this kind of division of labor, philosophy and physics can work in tandem to figure out whether the world is deterministic.

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